6,906: Economics: Macro for Computer Science

Lecture 3: Facts about Growth

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Learning objectives

- By the end of this lecture, you should be able to:
  - Describe the evolution of various economies over time based on facts of growth;

What is economic growth?

- Growth is the steady increase in aggregate output over time.
- Today we begin the study of economic growth and output determination in the long run.

Technicalities

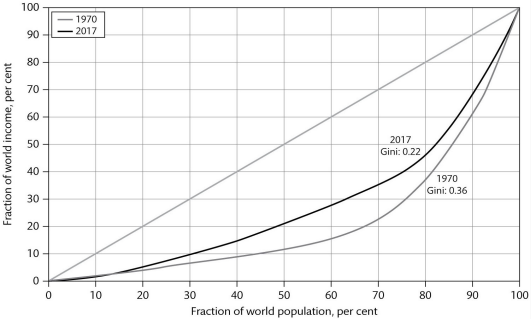
- We care about growth because we care about the standard of living.
- Today we will focus on economic performance across countries, which we measure in terms of real GDP per capita as it is indicative of average welfare across people.
- However, some economists prefer to use real GDP per worker for the comparison to account for the fact that some less developed countries have a larger informal sector and relatively more people living from non-marketed home production.
- Moreover, GDP per worker is a proxy for labor productivity and is thus preferred for the comparisons of productivity across countries.

Technicalities

- When comparing GDP across countries, a natural complication is that different countries have different currencies, and hence different units of GDP.
- Hence, we need to correct for variations in exchange rates and systematic differences in prices across countries.
- In the analysis which follows, we measure GDP across countries in terms of international dollars using a worldwide price index that accounts for cross-country differences in the purchasing power of different currencies.
- Details on the construction of purchasing power parities (PPP) are found in Blanchard (2018), Chapter 10.

Long run growth: Cross country stylized facts

- "Stylized" means these facts are roughly true over sufficiently large periods.
- **Stylized fact 1:** Some countries are rich and some are poor. However, there is some tendency towards a more equal world income distribution, though not much at the very bottom.



Source: Source: Sorensen and Whitta-Jacobsen (2022)

Long run growth: Cross country stylized facts

- **Stylized fact 1:** Some countries are rich and some are poor. However, there is some tendency towards a more equal world income distribution, though not much at the very bottom.

		GDP per Person
High income countries		
	Canada	$35,180
	Germany	$34,383
	Japan	$30,232
	Singapore	$59,149
	United Kingdom	$32,116
	United States	$42,426
Middle income countries		
	China	$8,640
	Dominican Republic	$8,694
	Mexico	$12,648
	South Africa	$10,831
	Thailand	$9,567
	Uruguay	$13,388
Low income countries		
	Cambodia	$2,607
	Chad	$2,350
	India	$3,719
	Kenya	$1636
	Mali	$1,157
	Nepal	$1,281

Notes: This data comes from the Penn World Tables, version 8.1. The real GDP is in terms of chain-weighted PPPs.

Source: Garin, Lester, and Sims (2018)

Long run growth: Cross country stylized facts

- The table shows considerable variations in income across countries based on the level of output per person in 2011 for a selected subset of countries.
- In purchasing power parity terms (PPPs), the average person in the United States was 36.67 times ($42,426/$1,157) richer than the average person in Mali. This is an enormous difference.
- In 2011, 29 countries had an income per capita of five percent or less than that in the U.S.
- Even among relatively rich countries, there are still important differences in output per capita. For example, in the US real GDP per capita is about 30 percent larger than it is in Great Britain and about 25 percent larger than in Germany.

Long run growth: Cross country stylized facts

- **Stylized fact 2:** Growth rates vary substantially between countries, and by the process of growing or declining fast, a country can move from being relatively poor to being relatively rich, or from being relatively rich to being relatively poor.

	Growth Miracles				Growth Disasters		
	1970 Income	2011 Income	% change		1970 Income	2011 Income	% change
South Korea	$1918	$27,870	1353	Madagascar	$1,321	$937	-29
Taiwan	$4,484	$33,187	640	Niger	$1,304	$651	-50
China	$1,107	$8,851	700	Burundi	$712	$612	-14
Botswana	$721	$14,787	1951	Central African Republic	$1,148	$762	-34

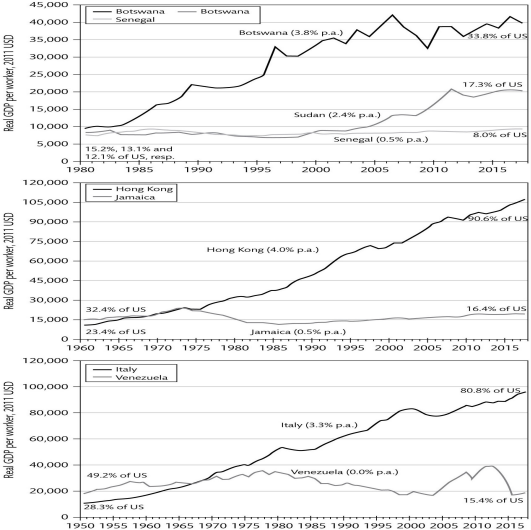
Notes: This data comes from the Penn World Tables, version 8.1. The real GDP is in terms of chain-weighted PPPs.

Source: Garin, Lester, and Sims (2018)

Long run growth: Cross country stylized facts

- **Stylized fact 3:** Growth can break in a country, turning from a high rate to a low one or vice versa.
- This implies that the fight against poverty is not hopeless as there are no growth-preventing areas in the world, but there may be growth-preventing policies or circumstances (e.g. war).

Long run growth: Cross country stylized facts

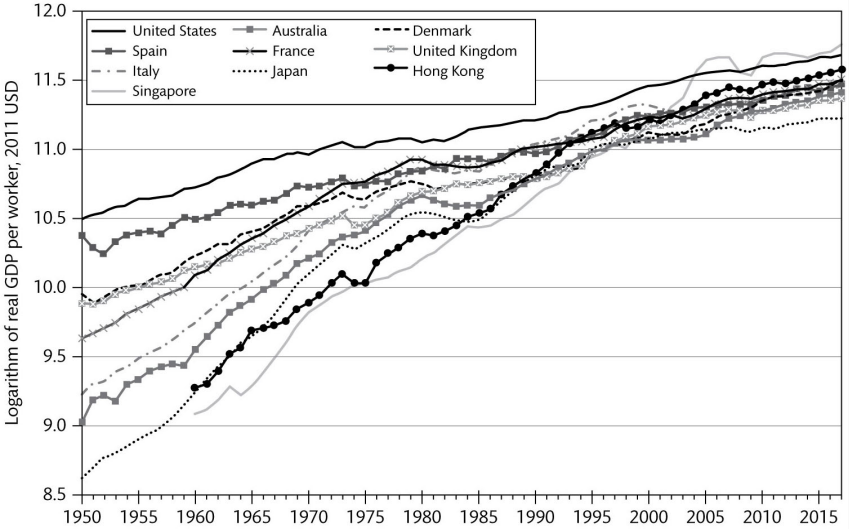


Source: Source: Sorensen and Whitta-Jacobsen (2022)

Long run growth: Cross country stylized facts

- **Stylized fact 4 (Convergence):** A country's income per worker (or capita) converges to a long-run growth path that depends on the country's basic structural characteristics and on whether its initial GDP per capita is above or below a specific threshold value. The further below the relevant growth path a country starts out, the faster it will grow.
- This implies that income per worker converges to the same level across countries conditional on the countries being structurally alike (e.g. investing similarly in physical or human capital) and on the countries starting on the same side of the threshold value.

Long run growth: Cross country stylized facts



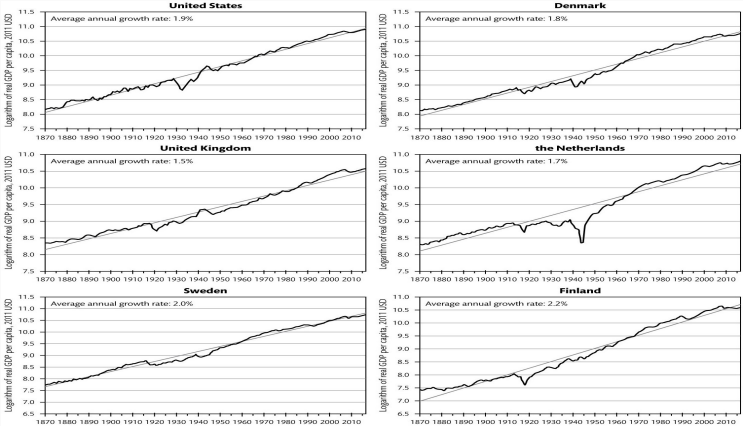
Source: Source: Sorensen and Whitta-Jacobsen (2022)

Long run growth: Cross country stylized facts

- For the OECD countries, there is clear evidence of convergence.
- Convergence is also visible in many Asian countries, especially for those with high growth rates, called the four tigers - Singapore, Taiwan, Hong Kong, and South Korea.
- Most African countries were poor in 1960, and some of them had negative growth of output per person between 1960 and 2017 due in part to internal or external conflicts.

Long run growth: Individual country stylized facts

- **Stylized fact 5:** Real GDP per worker grows at a sustained, roughly constant, rate over long periods (in advanced economies).



Source: Source: Sorensen and Whitta-Jacobsen (2022)

Long run growth: Cross country stylized facts

	Annual Growth Rate Output per Person (%)	Real Output per Person (2011 dollars)		
	1950–2017	1950	2017	2017/1950
France	2.6	7,025	39,461	5.6
Japan	4.1	2,531	40.374	15.9
United Kingdom	2.1	9,354	39,128	4.2
United States	2.0	14,569	54,995	3.8
Average	2.7	8,370	43,490	5.2

Source: Blanchard (2021) based on Penn World tables. The average in the last line is a simple unweighted average.

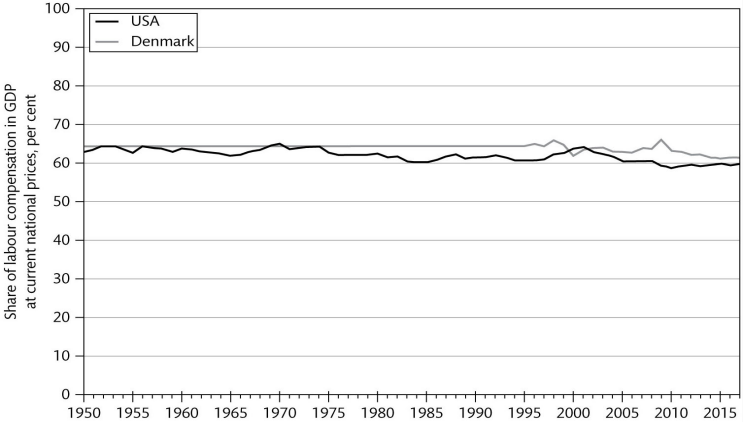
Long run growth: Individual country stylized facts

- **Stylized fact 6:** During the long periods of relatively constant growth rates in GDP per worker in the typical Western economy, labor's share of GDP has stayed relatively constant. Hence, the average real wage of a worker has grown by approximately the same rate as GDP per worker.

$$Laborshare = \frac{w_t N_t}{Y_t},$$

where w_t is the (real) wage, N_t is total labor input, and Y_t is output.

Labor's share of GDP



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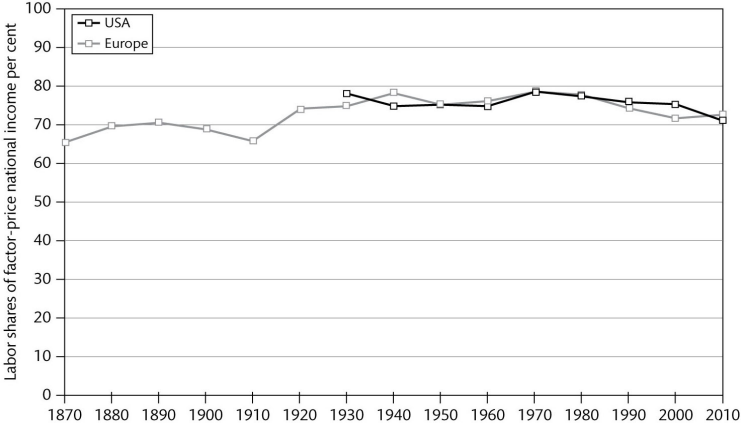
Labor's share of GDP



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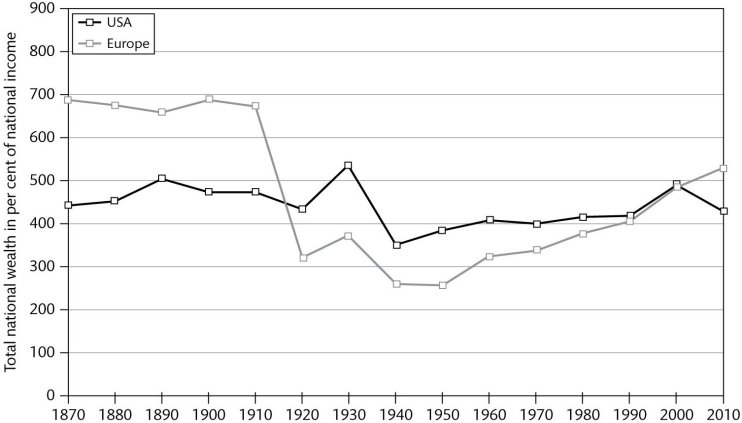
Long run growth: Individual country stylized facts

- **Stylized fact 7:** During the long periods of relatively constant growth rates in GDP per worker in the typical Western economy, capital's (wealth's) income share and the rate of return on capital have shown no trend, therefore the capital-output ratio K/Y has been relatively constant, and the capital intensity K/N has grown by approximately the same rate as GDP per worker.

$$Capitalshare = 1 - Laborshare = \frac{r_t K_t}{Y_t}$$

where r_t denotes the rental rate of capital and K_t the total stock of capital.

Wealth - income ratios



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Balanced growth

The growth process follows a balanced growth path if:

- GDP or income per worker, consumption per worker, the real wage rate, savings/investment per worker, and the capital - labor ratio all grow at one and the same constant rate, g.
- The labor force (population) grows at a constant rate, n.
- GDP, consumption, savings/investment, and capital grow at the common rate, $g + n$.
- The capital-output ratio and the rate of return on capital are constant.

Summary

- In this lecture we learned:
  - To identify the stylized facts that describe the differences in economic growth across countries.
- Next lecture we will study the macroeconomic model which allows us to analyze economic growth.